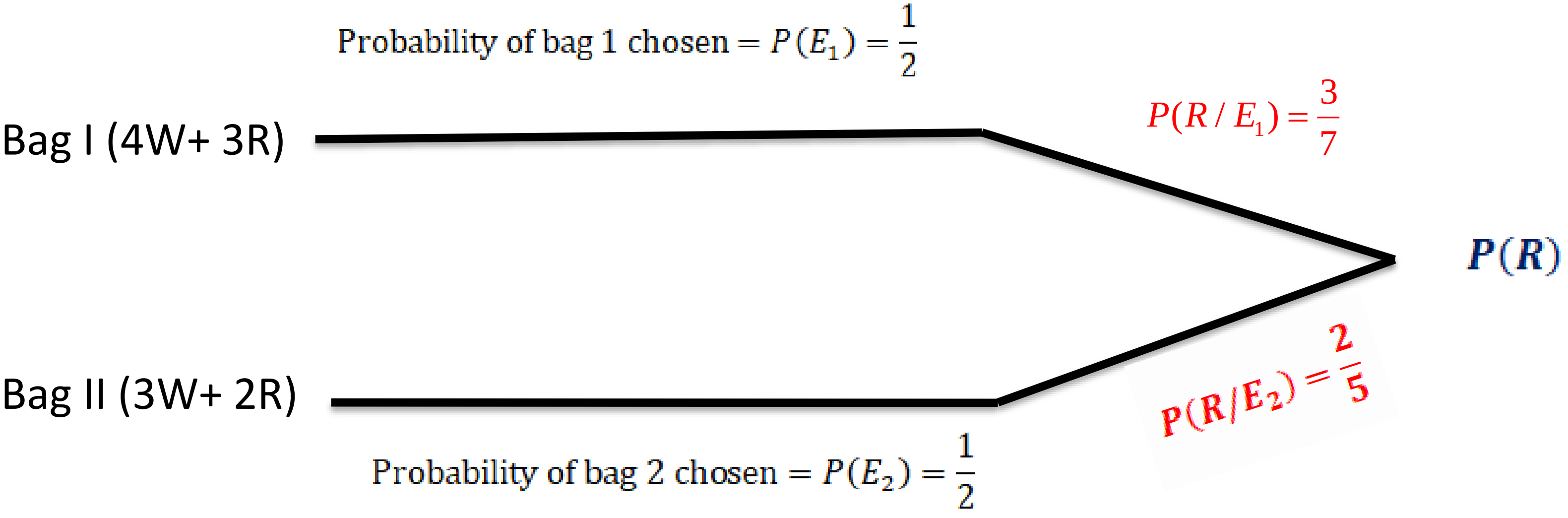
Bayes’ Theorem

There are two bags in front of you and you have remove one ball randomly what is the probability that ball is red ?



Total probability = $P(E_1) \times P(R/E_1) + P(E_2) \times P(R/E_2)$

Total probability $= \frac{1}{2} \times \frac{4}{7} + \frac{1}{2} + \frac{2}{5} = \frac{29}{70}$

Note that Events should be mutually exclusive

Lets us consider a new situation ,we know that ball we drew was red in color we looked at that now find the probability that ball is coming from bag II.
This type of event is called **reverse probability**

$$P(E_2/R) = \frac{P(E_2) \times P(A/E_2)}{P(A)} \qquad P(E_2/R) = \frac{\frac{1}{2} \times \frac{2}{5}}{\frac{29}{10}} = \frac{14}{25}$$

In a certain assembly plant, three machines, B1, B2, and B3, make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected. What is the probability that it is defective?

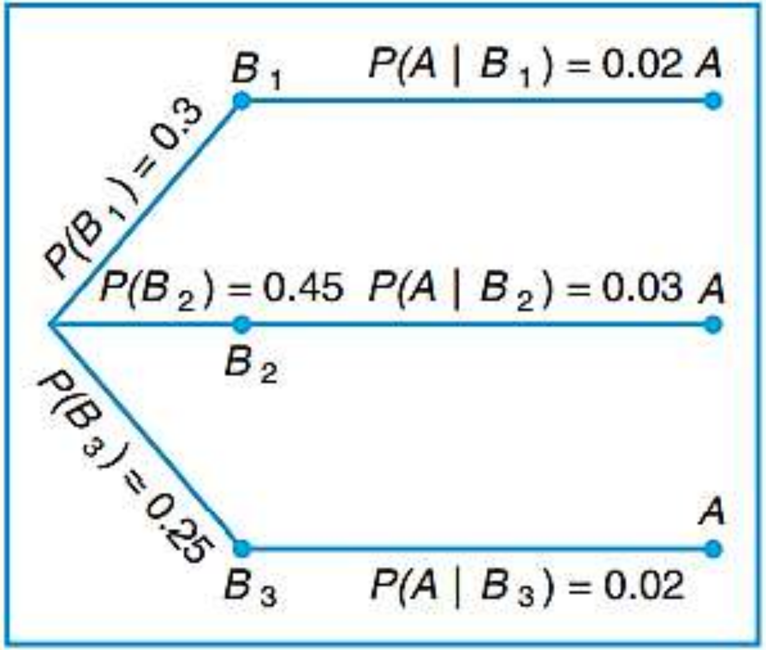
Solution: Consider the following events:

A: the product is defective,

B1: the product is made by machine B1, so $P(B1)=0.3$

B2: the product is made by machine B2, so $P(B2) = 0.45$

B3: the product is made by machine B3, so $P(B3)= 0.25$



The product is defective given it came from Machine 1 = $P(A/B1)$=0.02

The product is defective given it came from Machine 2 = $P(A/B2)$= 0.03

The product is defective given it came from Machine 3 = $P(A/B3) = 0.02$

$$P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3).$$

$$P(B_1)P(A|B_1) = (0.3)(0.02) = 0.006,$$
$$P(B_2)P(A|B_2) = (0.45)(0.03) = 0.0135,$$
$$P(B_3)P(A|B_3) = (0.25)(0.02) = 0.005,$$

P(A)=0.006 + 0.0135 + 0.005 = = 0.02

(B)

if a product was chosen randomly and found to be defective, what is the probability that it was made by machine B3?

***Solution*:** Using Bayes' rule to write

$$P(B_3|A) = \frac{P(B_3)P(A|B_3)}{P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3)},$$

$$P(B_3|A) = \frac{0.005}{0.006 + 0.0135 + 0.005} = \frac{0.005}{0.0245} = \frac{10}{49}.$$

QUESTION .

Customers are used to evaluate preliminary product designs. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and 25% have been poor products.

(a) What is the probability that a product attains a good review?

(b) If a new design attains a good review, what is the probability that it will be a highly successful product?

(c) If a product does not attain a good review, what is the probability that it will be a highly successful product

Let G denote a product that received a good review. Let H, M, and P denote products that were high, moderate, and poor performers, respectively.

a)

$$\begin{aligned} P(G) &= P(G|H)P(H) + P(G|M)P(M) + P(G|P)P(P) \\ &= 0.95(0.40) + 0.60(0.35) + 0.10(0.25) \\ &= 0.615 \end{aligned}$$

b) Using the result from part a.,

$$P(H|G) = \frac{P(G|H)P(H)}{P(G)} = \frac{0.95(0.40)}{0.615} = 0.618$$

c) $$P(H|G') = \frac{P(G'|H)P(H)}{P(G')} = \frac{0.05(0.40)}{1-0.615} = 0.052$$

EXAMPLE

The probability that the test correctly identifies someone with the illness as positive is 0.99, and the probability that the test correctly identifies someone without the illness as negative is 0.95. The incidence of the illness in the general population is 0.0001. You take the test, and the result is positive. What is the probability that you have the illness?